

About Simulation Input & Data Standard - (SIDS)

Simulation

Every simulation depends on inputs.

Those inputs may come from sensors, historical records, operational systems, scientific measurements, external models, human operators, AI systems, synthetic generators, or previous simulations.

But information does not become suitable simulation input merely because it exists or because a simulation can technically ingest it.

Simulation Input & Data Standard (SIDS) governs the conditions under which information may legitimately enter and influence a simulation.

Canonical Definition

Simulation Input & Data Standard (SIDS) defines the governance conditions under which data, values, signals, measurements, states, distributions, synthetic inputs, and other information may enter a simulation and influence its behavior, requiring material inputs to remain sufficiently identifiable, representative, transformation-governed, temporally appropriate, uncertainty-aware, and traceable to their simulation use.

Why SIDS Exists

A strong simulation model can still produce weak or misleading results when the information entering it is: unrepresentative, stale, temporally misaligned, transformed incorrectly, based on the wrong population, missing important variation, synthetically generated without disclosure, uncertain, incomplete, or incompatible with the simulation context.

SIDS therefore separates:

Data Availability

from:

Simulation Input Suitability

The key question is not:

Do we have data?

It is:

Is this the right information to enter this simulation under these conditions?

Governed Space

Simulation Inputs & Data

SIDS governs data specifically at the point where it becomes capable of influencing simulation behavior.

It does not attempt to govern the entire organizational data lifecycle.

Its focus is the simulation-input boundary.

Governance Flow

Governed Simulation Model & Environment

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Input Need

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Simulation Input Identification

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Representativeness Assessment

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Transformation Governance

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Temporal Qualification

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Input Uncertainty

↓

Acceptance / Restriction / Exclusion

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Governed Simulation Input & Data Basis

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Simulation Scenario Design What SIDS Governs

SIDS establishes whether material Simulation Inputs are:

Identifiable — what exactly entered the simulation?

Representative — do the inputs sufficiently represent the conditions the simulation is intended to model?

Transformation-Governed — what happened to the data before simulation use?

Temporally Appropriate — does the data represent the right point, sequence, period, or system state?

Uncertainty-Aware — what measurement, estimation, sampling, transformation, or representativeness uncertainty remains?

Traceable — which inputs influenced which simulation configuration and execution?

Representativeness Matters

Data can be accurate and still be inappropriate for a simulation.

For example:

data collected during normal operation may accurately describe normal behavior while being unsuitable for a simulation of emergency failure.

Likewise:

data from one population, climate, geographic region, system generation, or operating state may not represent another.

SIDS therefore treats representativeness as a simulation-specific governance question.

Representative for what?

must always remain connected to:

Simulation Object + Purpose + Scope + Reality Representation + Scenario
Context Transformation Must Remain Visible

Simulation inputs are frequently transformed before use.

They may be: normalized, aggregated, filtered, interpolated, extrapolated, encoded, converted, cleaned, derived, imputed, or synthetically generated.

These transformations may materially change the information entering the simulation.

SIDS therefore preserves the chain:

Source Data → Transformation → Simulation Input

A transformed input should not silently be represented as identical to the original source information.

Synthetic Inputs

Synthetic data can be essential to simulation.

It may enable testing of: rare failures, extreme environments, unavailable populations, hypothetical systems, future conditions, edge cases, or dangerous scenarios that cannot reasonably be reproduced in reality.

SIDS does not treat synthetic data as inferior by default.

It requires one essential distinction:

Synthetic Input ≠ Observed Reality

The synthetic origin and the conditions under which that input may be used must remain visible.

Temporal Context Matters

Simulation data always has a relationship to time.

SIDS therefore asks:

When was this information observed?

Which period does it represent?

Is it sufficiently current for this simulation?

Do multiple inputs represent the same point in time?

Does sequence matter?

Has temporal aggregation hidden material events?

This becomes especially important for fast-changing systems such as: AI, autonomous systems, financial systems, infrastructure, transportation, digital twins, or live operational simulations.

Uncertainty Must Travel With the Input

Measured or estimated inputs are rarely perfectly certain.

A sensor may have tolerance.

Historical data may have sampling limitations.

Synthetic data may depend upon generation assumptions.

Interpolated values may not have been directly observed.

SIDS requires material uncertainty to remain connected to the input rather than disappearing when the data enters the simulation.

The simulation therefore should not imply greater certainty than its input

basis supports.

Missing Data

Missing information is itself a governance condition.

SIDS prevents missing values from silently becoming:

zero

average

default

or:

synthetic replacement

without an identifiable treatment.

Possible responses may include substitution, imputation, scenario restriction, conditional use, or exclusion.

The important requirement is visibility.

Conflicting Inputs

Different sources may disagree.

For example:

Sensor A = 50

while:

Sensor B = 75

SIDS requires material conflict to remain identifiable and the treatment of that conflict to remain governable.

A simulation should not simply choose the more convenient value without preserving why that choice was made.

Input Sets

A simulation execution rarely depends on one input.

It consumes a complete Simulation Input Set.

SIDS allows each execution to remain connected to the exact set of materially relevant information that entered it.

This enables later reconstruction:

Which inputs produced this result?

Which were measured?

Which were synthetic?

Which had been transformed?

Which were estimated or uncertain?

Which represented an older system state?

Business Value

For organizations using simulation, SIDS provides direct operational value.

It helps detect situations where expensive simulation capability is being fed with information that is unsuitable for the question being simulated.

This may reveal: weak datasets, hidden population gaps, temporal mismatch, synthetic-data dependence, transformation loss, stale inputs, incompatible data sources, missing information, and outputs that appear precise despite uncertain input conditions.

SIDS therefore improves the ability to: compare simulation runs, explain outputs, reproduce results, challenge simulation conclusions, update simulations, trace failures, and prepare simulation outputs for downstream validation.

Relationship to Simulation Assumptions

SAS governs what the simulation assumes.

SIDS governs what information the simulation consumes.

An assumption may influence how an input is interpreted, but the two remain distinct.

For example:

Assumption: Weather remains stable.

Input: Measured temperature at simulation start.

Relationship to Model & Environment

SMES establishes the controlled model and environment.

SIDS determines what information may enter that governed structure.

The relationship is:

Model & Environment → Input Requirements → Governed Input & Data Basis
Relationship to Scenario Design

SIDS does not determine which scenarios should be run.

It determines which information is available and suitable to support those scenarios.

The next Parent Standard, Simulation Scenario Standard (SSS), governs:

Which combinations of conditions, variations, edge cases, stresses, and alternatives should actually be simulated?

Relationship to VALIDOS®

SIDS does not perform universal data validation.

VALIDOS® may separately validate: an input dataset, a source, an evidence object, a simulation output, or the simulation itself.

The boundary is:

SIDS governs whether information may function as simulation input.

VALIDOS® governs formal validation of the relevant Validation Object.

Relationship to ORA™

ORA™ governs actual Operational Reality.

Simulation Inputs may originate from that reality.

SIDS governs the transformation from:

Operational information

into:

governed simulation input

without redefining Operational Reality itself.

Relationship to INTEGROS® and ADIT®

INTEGROS® may govern integrity of input objects, records, transformations, and preserved states.

ADIT® may preserve evidence of input acquisition, transformation, substitution, and execution use.

SIDS remains focused on the simulation-specific suitability and use of the input.

Architectural Position

SIDS is the sixth Parent Standard of SIMULOS® — Simulation Governance Architecture.

The progression is:

Simulation Object



Simulation Purpose & Scope



Reality Representation



Simulation Assumptions



Simulation Model & Environment



Simulation Inputs & Data



Simulation Scenario Design

SIDS therefore marks a critical transition.

Before SIDS, SIMULOS® establishes:

what the simulation is, what it represents, what it assumes, and through which structure it will operate.

SIDS then establishes:

what information may legitimately enter that structure.

Only after that foundation exists can scenario governance meaningfully determine what conditions the simulation should be asked to explore.

Foundational Principle

Information becomes legitimate simulation input only when what it represents, how it was transformed, when it applies, what uncertainty it carries, and where it may influence the simulation remain sufficiently governed.

Canonical Closing Statement

Simulation Input & Data Standard (SIDS) governs the information boundary of SIMULOS® by ensuring that material inputs remain identifiable, sufficiently representative, transformation-governed, temporally appropriate, uncertainty-aware, and traceable to the simulation activity they influence. By distinguishing technical ingestibility from methodological suitability, observed information from synthetic input, current information from temporally

mismatched data, and raw information from transformed input, SIDS provides the governed Input & Data Basis required before simulation conditions can be intentionally organized into scenarios and executed.

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